

An Unsupervised Heuristic-Based Hierarchical Method for Name Disambiguation in Bibliographic Citations

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Name ambiguity in the context of bibliographic citations is a difficult problem which, despite the many efforts from the research community, still has a lot of room for improvement. In this article, we present a heuristic-based hierarchical clustering method to deal with this problem. The method successively fuses clusters of citations of similar author names based on several heuristics and similarity measures on the components of the citations (e.g., coauthor names, work title, and publication venue title). During the disambiguation task, the information about fused clusters is aggregated providing more information for the next round of fusion. In order to demonstrate the effectiveness of our method, we ran a series of experiments in two different collections extracted from real-world digital libraries and compared it, under two metrics, with four *representative* methods described in the literature. We present comparisons of results using each considered attribute separately (i.e., coauthor names, work title, and publication venue title) with the author name attribute and using all attributes together. These results show that our unsupervised method, when using all attributes, performs competitively against all other methods, under both metrics, losing only in one case against a supervised method, whose result was very close to ours. Moreover, such results are achieved without the burden of any training and without using any privileged information such as knowing a priori the correct number of clusters.

Introduction

Digital Libraries (DLs) are complex information systems that involve rich sets of digital objects and their respective metadata, along with multiple organizational structures and services such as searching, browsing, and personalization, and they are normally built having a target community of users with specific interests (Gonçalves, Fox, Watson, & Kipp, 2004).

DLs, such as DBLP,¹ CiteSeer,² MEDLINE,³ and BDBComp,⁴ have become an important source of information for academic communities since they allow the search and discovery of relevant publications in a centralized way. Also, studies about the DL content can lead to interesting results such as coverage of topics, tendencies, quality and impact of publications of a specific sub-community or individuals, patterns of collaboration in social networks, etc. These types of analysis and information, which are used, for instance, by funding agencies to make decisions on grants and for an individual's promotion, presuppose *high-quality content*.

Citations (here regarded as a set of bibliographic attributes such as author and coauthor names, work title, publication venue title, etc.)⁵ are an essential component of these DLs (Lee, Kang, Mitra Giles, & On, 2007). Citation management within DLs involves a number of tasks. One in particular, name disambiguation, has received a lot of attention from the DL research community due to its inherent difficulty. Name ambiguity in the context of citations occurs when one author can be referred to by multiple name variations or when two or more people have exactly the same name or share a same name variation. This problem may occur due to number of reasons, including the lack of standards and common practices and the decentralized generation of content, e.g., by means of automatic harvesting (Lagoze & de Sompel, 2001).

More specifically, the name ambiguity problem, as seen in the context of bibliographic citations, can be divided into two sub-problems (Lee, On, Kang, & Park, 2005): *split citation* (SC) and *mixed citation* (MC). The former consists of knowing whether the citation records for an author are split into several distinct clusters associated with different variations

¹<http://dblp.uni-trier.de>

²<http://citeseer.ist.psu.edu>

³<http://medline.cos.com/>

⁴<http://www.lbd.dcc.ufmg.br/bdbcomp>

⁵Notice that in here we are not considering any other information that does not appear in citations such as affiliation, keywords, abstract, grants, acknowledgments, etc.

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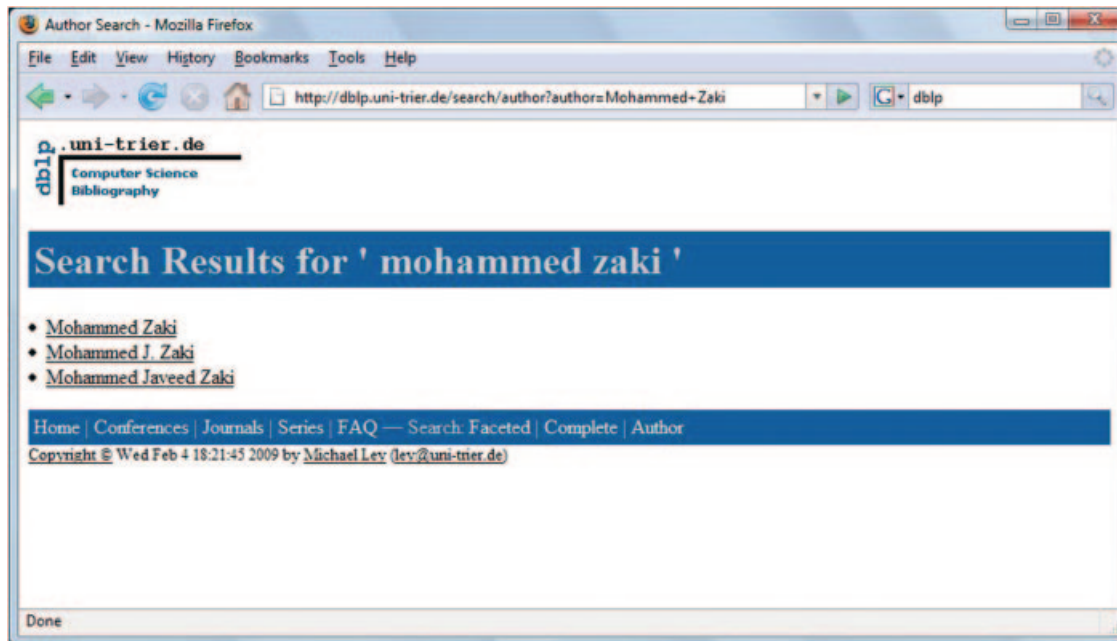


FIG. 1. Split citation: A unique author with several name variations.

of an author's name, as if these variations corresponded to different people. This occurs mainly due to misspelling errors, abbreviations, change in the order of name components (e.g., first and last names), among other reasons (Ley, 2002). A case of SC can be illustrated by searching DBLP for the author "Mohammed Zaki." In this case, the retrieved citations may contain three name variations—"Mohammed Zaki," "Mohammed J. Zaki," and "Mohammed Javeed Zaki" (Figure 1). Although all these three names may refer to the same person, they appear as if there were three distinct authors.

MC occurs when different authors share a common name and their publications appear as belonging to a unique single author only. An example is given in Figure 2. In that example, the name "Mohammed Zaki" refers to: Mohammed Zaki from Al-Azhar University, Nasr City, Cairo, Egypt. However, the second citation corresponds to an article coauthored by Mohammed Zaki from the Department of Computer Science, Rensselaer Polytechnic Institute, USA. There are many other cases in which two different authors simply have the same name, a common situation, for example, for authors with Asian names.

The complexity of dealing with this problem has led to a myriad of proposals of methods and approaches for its solution (Bhattacharya & Getoor, 2006, 2007; Culotta, Kanani, Hall, Wick, & McCallum, 2007; Han, Giles, Zha, Li, & Tsioutsoulis, 2004; Han, Xu, Zha, & Giles, 2005; Han, Zha, & Giles, 2005; Huang, Ertekin, & Giles, 2006; Kang et al., 2009; Lee et al., 2005; Malin, 2005; On, Lee, Kang, & Mitra, 2005; On, Elmacioglu, Lee, Kang, & Pei, 2006; On & Lee, 2007; Soler, 2007; Song, Huang, Councill, Li, & Giles, 2007; Torvik, Weeber, Swanson, & Smalheiser, 2005; Torvik & Smalheiser, 2009; Treeratpituk & Giles, 2009). However, despite all these efforts, there is still a lot of room

for improvement of the current solutions. Specifically Cota, Gonçalves, and Laender (2007), proposed a heuristic-based hierarchical clustering (HHC) method to deal with the name ambiguity problem. The method successively fuses clusters of citations of similar author names based on several heuristics on the components of the citations. During the disambiguation task, the information about fused clusters is aggregated (e.g., all words in the titles are grouped together) providing more information for the next round of fusion. In that work, we evaluated it under only one metric and using two simple baselines, not specifically designed for the problem. In this article, we greatly extend our preliminary work (Cota et al., 2007) evaluating it by using collections of citations extracted from two real-world DLs, namely, DBLP and BDBComp, the same set of evidence (author and coauthor names, work title, and publication venue title), and two clustering metrics for their comparison. As baselines, we used two supervised (Han et al., 2004) and two unsupervised (Han et al., 2005, 2006) methods that can be considered as representative for the problem at hand, since they explore different paradigms and exploit, exactly like us, only the most common evidence present in bibliographic citations. In summary, the main contributions of this work are:

1. A detailed description of a new unsupervised hierarchical clustering method for name disambiguation in bibliographic citations that includes specific heuristics specially designed for the problem and that does not make use of any training or privileged information; and
2. A comparative evaluation of this method with representative methods for name disambiguation under the same conditions (same collections, same metrics, and same attributes). Such a comparative evaluation is missing in the literature for this specific problem and is an original contribution of this work. In particular, it shows that

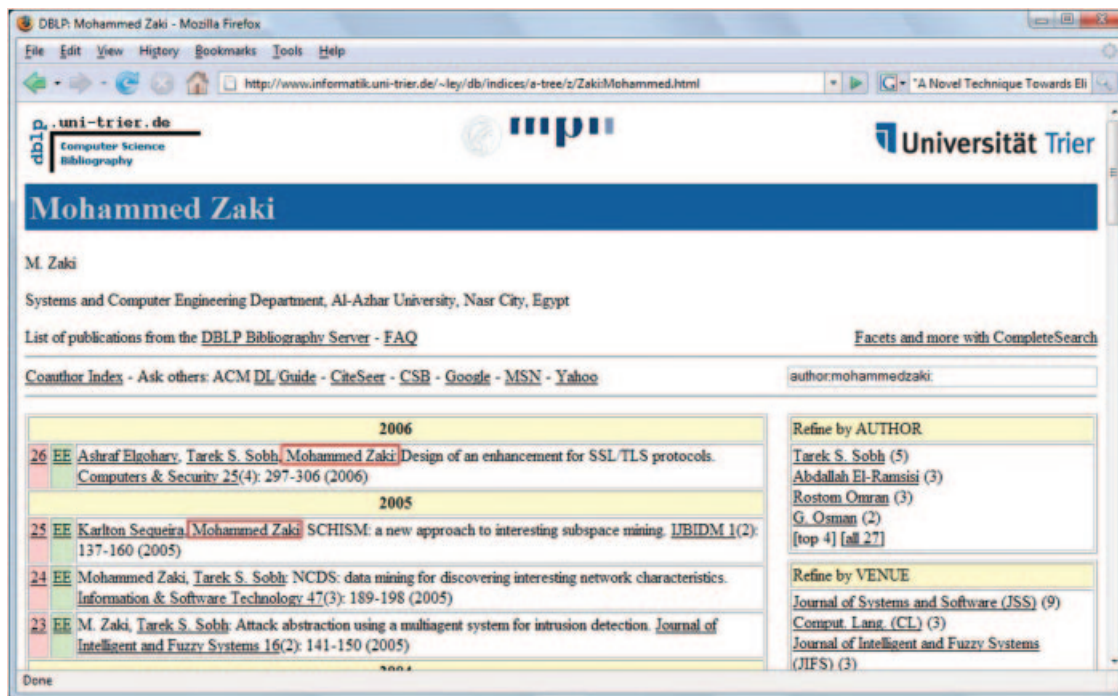


FIG. 2. Mixed citation: several authors with a same name variation.

our HHC method, when using all attributes, performs better than all the other methods under all metrics, except in two specific cases for which there is a statistical tie and a very close result to a supervised method. The absolute results also show that practical solutions that exploit only the most common information present in bibliographic citations can be devised. Moreover, these results are achieved without the burden of any training and without using any privileged information, such as knowing a priori the correct number of clusters.

The remainder of this article is organized as follows. The second section reviews related work. The third section presents the details of the proposed method. The section Comparative Evaluation describes the experimental setup and presents the results of the comparative evaluation. The fifth section analyzes some cases in which our method fails. The final section concludes the article along with a discussion on future work.

Related Work

The name disambiguation methods proposed in the literature adopt a wide spectrum of solutions (Smalheiser & Torvik, 2009) that range from those based on supervised learning techniques (Han et al., 2004) to those that use some unsupervised or semi-supervised clustering strategy (Bhattacharya & Getoor, 2006, 2007; Culotta et al., 2007; Han, Xu, et al., 2005; Han et al., 2005; Huang et al., 2006; On et al., 2005; Song et al., 2007; Torvik et al., 2005) or follow a graph-oriented approach (Malin, 2005; On et al., 2006; On and Lee, 2007). In this section, we present a brief review of some representative

name disambiguation methods. Our main focus, however, is on those methods that have been specifically designed to address the name disambiguation problem in the context of bibliographic citations, since they are more related to the scope of our work.

Being among the first to address the problem, Han et al. (2004) propose two methods based on supervised learning techniques that use coauthor names, work titles, and publication venues as evidence for name disambiguation. The first method uses a naive Bayes model, a generative statistical model frequently used in word sense disambiguation tasks, to capture all writing patterns in the authors' citations. The second method is based on support vector machines (SVMs), which are discriminative models basically used as a classifier (Mitchell, 1997). A basic difference between the two techniques is that a naive Bayes model requires only positive examples to learn about the writing patterns, whereas SVMs require both positive and negative examples to learn how to classify citations. Both methods have been evaluated with data taken from two datasets, one collected from the Web, mainly publication lists from homepages, and the other collected from DBLP. Experimental results show that, on average, using all attributes, the SVM-based method was more accurate than the naive Bayes-based method for the Web collected dataset, whereas for the DBLP dataset the naive Bayes-based method performed better. An unsupervised hierarchical version of the naive Bayes-based method is presented by Han, Xu, et al. (2005).

Han et al. (2005) propose an unsupervised learning method for name disambiguation that uses K-way spectral clustering, a graph model that has been successfully applied to data mining and cluster analysis. This method is also

evaluated with Web and DBLP datasets. Experimental results show that this method has achieved 63% of accuracy with the DBLP datasets, and up to 84.3% of accuracy with the Web datasets.

Huang et al. (2006) present a framework for solving the name ambiguity problem in which a blocking method first creates candidate classes of authors with similar names and then DBSCAN, a density-based clustering method (Ester, Kriegel, Sander, & Xu, 1996), is used for clustering citation records by authors. For each block, the distance metric between pairwise citations used in DBSCAN is calculated by an online active selection support vector machine algorithm (LASVM), which yields, according to the authors, a simpler model than the standard SVMs. For demonstrating the effectiveness of this framework, the authors have applied it to a manually annotated dataset with 3,335 citation records and 490 distinct authors obtaining 90.6% in terms of F1 when the evaluation is done to pairs of citations. It should be noticed that these results were obtained by exploiting additional sources of evidence such as information extracted from the headers of articles corresponding to the respective citations obtained from CiteSeer.

On et al. (2005) present a comparative study of alternative strategies for name disambiguation based on a two-step framework. In the first step (blocking), similar names are blocked together in order to reduce the number of candidates for the second step (disambiguation), which uses coauthor information to measure the distance between two names. Combining four blocking methods (initial first name and full last name comparison, token comparison, N-gram, and sampling-based join approximation) and seven distance metrics (naive Bayes and SVM-based metrics, Jaccard coefficient, Term Frequency—Inverse Document Frequency (TFIDF), Jaro, Jaro—Winkler, and cosine distance) on four datasets derived from DBLP, the authors present experimental results that allow identifying combinations that are scalable and effective to disambiguate author names in bibliographic citations.

On et al. (2006) propose a graph-based method for name disambiguation that uses a coauthorship graph in which each vertex represents an author name and each edge represents a coauthorship between two authors. This method also requires a preliminary step in which similar author names are clustered together (On et al., 2005). Thus, after these author contexts are determined and represented by a graph, the method finds the common quasi-clique between two vertices and the vertex degree of this quasi-clique is used as a similarity measure among their respective author names. Experimental results expressed in terms of average ranked precision measures, considering real and synthetic datasets, have shown that this method considerably improves the performance of traditional text similarity measures.

On and Lee (2007) study the scalability issue regarding the name disambiguation task. They examine two specific methods, K-way spectral clustering (Han et al., 2005) and multi-way distributional clustering (Bekkerman & McCallum, 2005), and point out their limitations with respect

to scalability. Then, using datasets taken from ACM DL, and DBLP, they show that a method based on the multi-level graph partition technique (Dhillon, Guan, & Kulis, 2005) might be successfully applied to solve the name disambiguation in large collections.

Torvik et al. (2005) propose a probabilistic metric for determining the similarity between MEDLINE records. The learning model is created using similarity profiles between articles. A similarity profile is a comparison vector between a pair of articles, used to indicate the similarity between them, based on the following attributes: work title, publication venue title, coauthor and author names, medical subject headings, language, and affiliation. The authors also propose some heuristics for generating training sets (positive and negative) automatically. When the probabilistic metric receives two records, their similarity profile is computed and the relative frequency of this profile in the positive and negative training sets is checked for determining whether these two records are authored by the same author or not. In a subsequent work, Torvik and Smalheiser (2009) extend this method including addition of features, new ways of automatically generating training sets, an improved algorithm for correcting the transitivity problem, and a new agglomerative clustering algorithm.

Bhattacharya and Getoor (2006) extend the Latent Dirichlet Allocation model and propose a probabilistic model for collective entity resolution that uses the cooccurrence of the references to entities⁶ in each work to determine the entities jointly, i.e., they use the disambiguated references to disambiguate other references in the same work. An algorithm for collective entity resolution that uses the attributes and relational information about the citation records is proposed by Bhattacharya and Getoor (2007).

Culotta et al. (2007) propose a more generic representation for the author disambiguation task that considers features over sets of records, instead of only features between pairs of records, and present a training algorithm that is error-driven, i.e., training examples are generated from incorrect predictions in the training data, and rank-based, i.e., the classifier provides a ranked result for the disambiguation. Kanani, McCallum, and Pal (2007), on the other hand, present several methods for increasing the author coreference by gathering additional evidence from the Web.

Song et al. (2007) propose a two-step unsupervised method. The first step uses Probabilistic Latent Semantic Analysis and Latent Dirichlet Allocation to assign a vector of probabilities of topics to each citation. In the second step, it considers the probability distribution of topics with respect to person names as new evidence for name disambiguation. et Allocation to assign a vector of probabilities of topics to a name. In the second step, they consider the probability distribution of topics with respect to person names as new evidence for name disambiguation.

Soler (2007) proposes a distance metric between two publications, based on terms and author names in common

⁶Here each author name in a work corresponds to a reference to an author.

between them, and also between two clusters of articles. In that work, the author proposes a semi-automatic algorithm that creates clusters of articles using the proposed distances and summarizes the clusters by means of a representative article of the cluster including the distance from it to other clusters. Then, the user chooses from among the several clusters the one that is most probable to belong to the same author.

Yang, Peng, Jiang, Lee, and Ho (2008) address the author name ambiguity problem in bibliographic citations using topics and correlations found on the Web. The authors determine the topics of the citation from venue information using an extraction algorithm based on association rules in order to create a topic association network. They also use the Web for retrieving publication pages of authors or coauthors to be disambiguated. Then, they create a similarity function making use of a binary classifier on top of all these features. This function is used by an algorithm to group citations of a same author. They tested their approach on the collection constructed by Han et al. (2004) and obtained an increase of the accuracy around 66% compared with the use of the citations without the topics and Web correlations.

Kang et al. (2009) explore coauthorship information using a Web-based technique that obtains implicit coauthors of the author to be disambiguated. They submit as query a pair of citation author names to Web search engines to retrieve documents containing both author names, and then extract new names found in these documents as new implicit coauthors of the pair.

Pereira et al. (2009) also exploit the Web for obtaining additional information to disambiguate author names. The proposed method attempts to find Web documents corresponding to curricula vitae or Web pages containing publications of a single author. If two citations of two ambiguous authors occur in the same Web document, these citations are considered as belonging to the same author and are fused in the same cluster. One problem with this method and the previous one is the additional cost of extracting all the needed information from Web documents.

Finally, Treeratpituk and Giles (2009) propose a pairwise linkage function for author name disambiguation in the MEDLINE DL. The authors exploit a large feature set obtained from MEDLINE metadata, similar to that of Torvik et al. (2005), and assess the effectiveness of random forests, in comparison to other classifiers, for constructing a pairwise linkage function to be used along with some author name disambiguation algorithms (left unspecified). They also investigate the performance of subsets of the features capable of reaching good effectiveness.

Our method, in contrast to the methods described above, considers only the most commonly used citation features, does not exploit any privileged information such as the number of corrected classes/clusters to be generated and does not need a training set to create a (effective) model. Our experiments show that HHC presents competitive results when using all attributes, without the additional burden

of any training process and without using any privileged information.

Since the name ambiguity problem is not restricted to a single context, it is worth noting that several other name disambiguation methods, which exploit distinct pieces of evidence or are targeted to other applications (e.g., name disambiguation in Web search results), have been described in the literature. For instance, Malin (2005) proposes two methods for name disambiguation that exploits the existing relations among ambiguous names. The first method is based on a hierarchical clustering strategy and the second one makes use of social networks. Vu et al. (2007) propose the use of Web directories as a knowledge base to disambiguate personal names in Web search results, whereas Bekkerman and McCallum (2005) present two methods for addressing this same problem, one based on the link structure of the Web pages and the other one using agglomerative/conglomerative double clustering, a multi-way distributional clustering. Galvez and de Moya-Anegón (2007) address the problem of conflating personal name variants in a canonical form using binary matrices and finite-state graphs. A more detailed discussion of these methods is out of the scope of this article.

The HHC Method

Our proposed disambiguation method deals at the same time with the SC and MC problems by combining similarity functions with some heuristics that use the evidence present in the citation records of authors that we want to disambiguate. By considering ascandidates for clustering, in the first step of our method, only records of *similar* authors (i.e., authors whose names have some similarity according to some matching function) and forcing them to have at least a common similar coauthor, we are dealing with the MC problem, i.e., we are diminishing the chances that the publication of two different authors will be mixed in the same group. On the other hand, by relying on other similarity evidence (but still considering similar names) based on the intuition that the publications of an author share some similar features (e.g., they have a common coauthor, they are about the same subject) to decide whether we should fuse already existing groups, we are dealing with the SC problem, i.e., we are joining together groups that should not be separated. At the same time, citation records of groups that do not share any similar features are not clustered together, thus diminishing the chances of mixing publications of different authors. Notice that solving these two problems at the same time is difficult since these two goals are in some sense in conflict and a good solution should find the proper balance.

Before we continue to describe the method, we will introduce some definitions that will be used hereafter.

Definition 1. (Citation Record). A citation record c is a set of bibliographic data, such as author names, work title, publication venue title, publication year, etc., that is pertinent to a particular article.

Definition 2. (Authorship Record). An authorship record represents the participation of an author in a citation record c . Thus, if a citation record c has n authors, it generates n authorship records a_i (one for each author). In this work, we consider that an authorship record is composed of four main attributes: author name, a list of coauthor names, work title, and publication venue title. In addition, it also includes a unique identifier for the citation record and the position of the author name within that record.

Definition 3. (Short Name). A short name is an author name composed of the initial of its first name and the full last name, such as, for example, “A. Gupta,” “M. Zaki,” and “J. Ullman.”

The HHC method attempts to resolve the name ambiguity problem in two main steps after a preprocessing phase, in which it treats citation records by removing stopwords and stemming the words of work and publication venue titles, and then creates authorship records and groups of authorship records that include ambiguous author names. Hereafter, we refer to these groups as *ambiguous groups*.

Algorithm 1 describes the HHC method. It receives a list R of citation records and returns a list C of clusters of authorship records. Lines 5–7 of the algorithm correspond to the preprocessing phase, creation of the authorship records, and creation of the ambiguous groups. To obtain the list G of ambiguous groups, any blocking strategy (On et al., 2005) may be used, since the main aim here is to reduce the complexity of the problem, since fusing only clusters with similar names considerably diminishes the number of possible comparisons. After the list G of ambiguous groups is produced, for each ambiguous group g in G , HHC applies its two main steps (lines 10 and 11) to generate the list of authorship record clusters (lines 7 and 12).

Algorithm 1. HHC.

Input: List R of citation records;

Output: List C of clusters of authorship records;

```

1  Let  $A$  be a list of authorship records;
2  Let  $C_1$  and  $C_2$  be lists of clusters;
3  Let  $G$  be a list of ambiguous groups;
4  Let  $R'$  be a list of citation records;
5   $R' \leftarrow \text{PreprocessCitationRecords}(R)$ ;
6   $A \leftarrow \text{CreateAuthorshipRecords}(R')$ ;
7   $G \leftarrow \text{CreateAmbiguousGroups}(A)$ ;
8   $C \leftarrow \emptyset$ 
9  for each ambiguous group  $g$  in  $G$  do
10      $C_1 \leftarrow \text{FirstStep}(g)$ ;
11      $C_2 \leftarrow \text{SecondStep}(C_1)$ ;
12      $C \leftarrow C \cup C_2$ ;
13 end for
```

Algorithm 2 describes the first HHC step. We will refer to Figure 3 when discussing it. This algorithm receives an ambiguous group g and returns a list C of clusters of authorship records. It processes g by splitting it into two separate lists: S with records whose author names occur in a short format (Figure 3a) and L with the remaining ones (i.e., those records whose ambiguous author names are not in a short format). Then, it proceeds by first processing L (line 6) and then S (line 7). When processing the lists L and S , the initial clusters of authorship records are built using the author name and the list of coauthor names as evidence. The idea of first processing the list of long names is that these names provide more evidence for our similarity functions to decide whether two authors have similar names or not. Figure 3b illustrates this processing applied to a sample list of authorship records S containing only short names. Note that the second and the third authorship records have been clustered together because they have in common a similar coauthor name (“Riccardo Bettati” and “Bettati, R.”). We have based this strategy on a general heuristic that assumes that very rarely two authors with similar names that share a coauthor in common would be two different people in the real world. As it will be seen in the experimental results, this heuristic proved to be very effective. This process generates very pure (i.e., with few wrong records within the cluster) but very fragmented (i.e., many records related to the same author are spread in different clusters) sets of clusters.

Algorithm 2. FirstStep.

Input: Ambiguous group g ;

Output: List C of clusters of authorship records;

```

1  Let  $L$  and  $S$  be lists of authorship records;
2  Let  $C$ ,  $C_1$  and  $C_2$  be lists of clusters;
3   $S \leftarrow \text{GetShortNameRecords}(g)$ ;
4   $L \leftarrow \text{GetLongNameRecords}(g)$ ;
5   $C_1 \leftarrow \emptyset$ ;
6   $C_2 \leftarrow \text{ProcessList}(L, C_1)$ ;
7   $C \leftarrow \text{ProcessList}(S, C_2)$ 
```

Algorithm 3 describes the function `ProcessList` used to process the list of authorship records in Algorithm 2 (lines 6 and 7). This algorithm receives a list A of authorship records and a list C_i of clusters of authorship records and returns a new list C_o with each authorship record a from A in some cluster c of C_o . It compares the author name from each authorship record a with the author name from the first authorship record within each cluster c using a similarity function. If the author name from a is similar to the author name from the first authorship record of c and there is a coauthor name in a that is similar to some coauthor name in c , a is inserted in this cluster c (line 7); otherwise a new cluster is created with this authorship record a (line 13), and is appended to the list C_o .

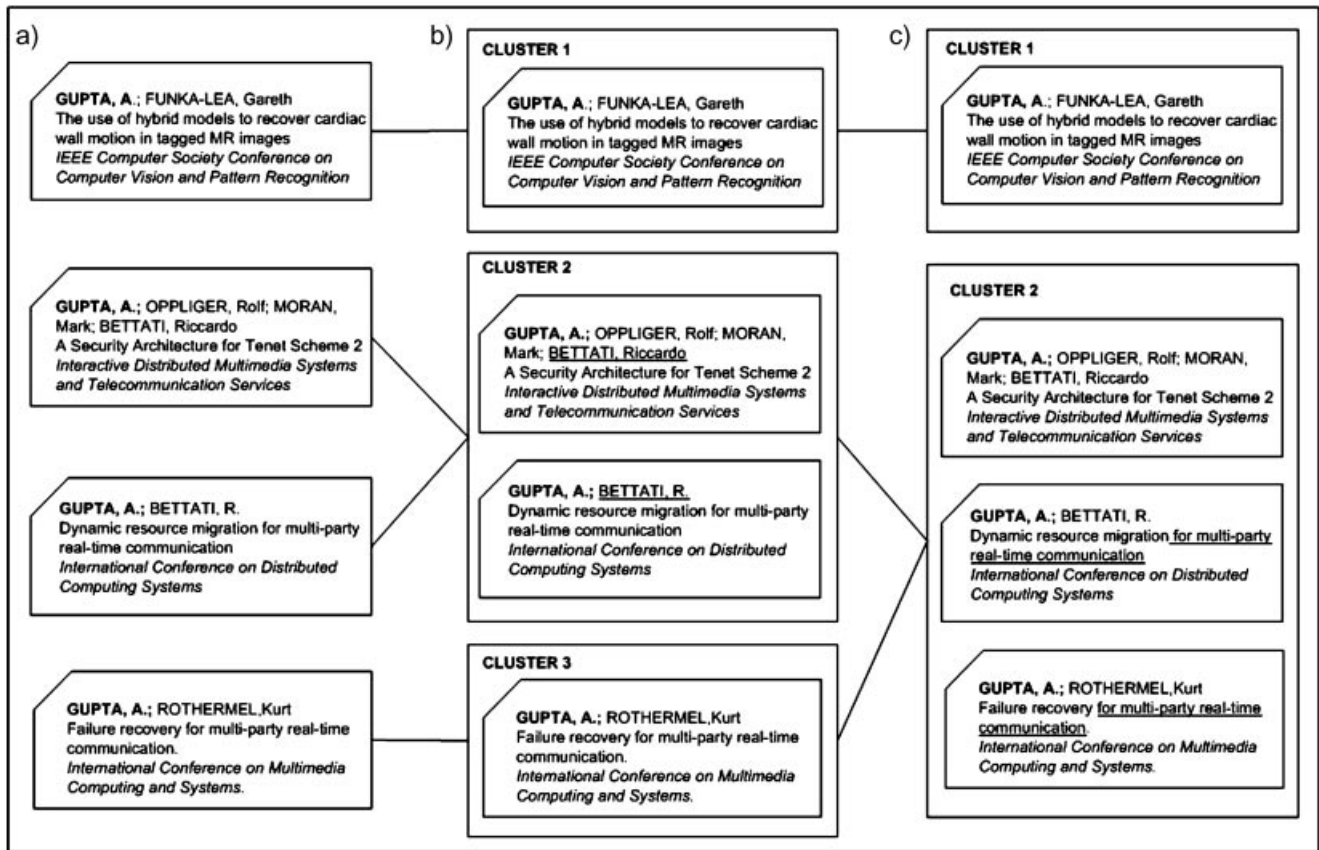


FIG. 3. (a) Initial list of authorship records. (b) Clusters generated in the first step. (c) Final result after the fusion step.

Algorithm 3. ProcessList function.

Input: List A of authorship records;
Input: List C_i of authorship record clusters;
Output: List C_o of authorship record clusters;

```

1   $C_o \leftarrow C_i$ ;
2  for each  $a$  in  $A$  do
3       $inserted \leftarrow \text{false}$ ;
4       $c \leftarrow \text{first}(C_o)$ ;
5      while not  $inserted$  and  $c \neq \text{null}$  do
6          if the author name from  $a$  is similar to the
              author name from the first authorship
              record of  $c$  and there is a coauthor name
              in  $a$  that is similar to some coauthor
              name in  $c$  then
7              InsertAuthorshipRecord( $a, c$ );
8               $inserted \leftarrow \text{true}$ ;
9          end if
10          $c \leftarrow \text{next}(C_o)$ ;
11     end while
12     if  $inserted = \text{false}$  then
13          $c \leftarrow \text{CreateNewCluster}(a)$ ;
14         Append( $C_o, c$ );
15     end if
16 end for

```

Algorithm 4 describes the second HHC step. This step tries to reduce fragmentation by using additional information from the authorship records in the initial clusters (in our implementation we used work and publication venue titles) in order to fuse them. This is based on the general heuristic that presupposes that authors tend to publish on the same subjects and venues for some portion of their careers. Thus, in the second step, HHC proceeds in a hierarchical manner and fuses the clusters created in step one based on their contents. Clusters are pairwise compared (line 12) and, to determine whether two clusters can be fused, HHC estimates their similarity based on the elements that compose the work and publication venue title of their respective citations, if the names of the authors of the clusters are similar. If the estimated similarity of the elements of either these attributes is higher than a given threshold, the two clusters are fused (lines from 12 to 16). This process continues until the termination criterion is met, i.e., until no more clusters can be fused. The final result is a list C_o of clusters with their respective authorship records (Figure 3c).

Notice that each time two or more clusters of similar authors are fused, the resulting cluster becomes “information richer” since the information about fused clusters is aggregated (e.g., all words in the publication titles are considered together) providing more information for the next round of fusion. This helps to increase the similarity

Algorithm 4. SecondStep.

Input: List C_i of clusters of authorship records;

Output: List C_o of clusters of authorship records;

```
1   $C_o \leftarrow C_i$ ;  
2   $fused \leftarrow \text{true}$ ;  
3  while  $fused$  do  
4     $fused \leftarrow \text{false}$ ;  
5    for each  $c_1$  in  $C_o$  do  
6      for each  $c_2$  in  $C_o$  do  
7        if  $c_1 \neq c_2$  and the first author name from  $c_1$  is  
          similar to the first author name from  $c_2$  then  
8           $t_{t1} \leftarrow \text{GetWorkTitleTerms}(c_1)$ ;  
9           $t_{t2} \leftarrow \text{GetWorkTitleTerms}(c_2)$ ;  
10          $t_{v1} \leftarrow \text{GetPublicationVenueTitleTerms}(c_1)$ ;  
11          $t_{v2} \leftarrow \text{GetPublicationVenueTitleTerms}(c_2)$ ;  
12         if  $\text{TitleSimilarity}(t_{t1}, t_{t2}) > \text{title-threshold}$   
           or  $\text{VenueSimilarity}(t_{v1}, t_{v2}) > \text{venue-}$   
              $\text{threshold}$   
           then  
13              $c_1 \leftarrow \text{Fuse}(c_1, c_2)$ ;  
14              $\text{remove}(C_o, c_2)$ ;  
15              $fused \leftarrow \text{true}$ ;  
16           end if  
17         end if  
18       end for  
19     end for  
20   end while
```

among clusters, which is recomputed after each fusing iteration.

To avoid recomputing the pairwise similarity measure between the clusters, we can use a priority queue or a lookup table, as proposed by Smalheiser and Torvik (2009). Using a priority queue, each position in this queue would contain, besides the clusters' identifier, two fields storing the work title and publication venue title similarities between two clusters. Another data structure would be necessary to indicate, given a cluster identifier, which positions of the queue refer to this cluster. This would be used when two clusters are fused and we would need removing or updating the similarity measures on the original clusters from the queue.

Different similarity functions can be used for each kind of attribute. In our experiments, for author and coauthor names, we used a function derived from the Fragment Comparison algorithm, an edit-distance matching algorithm specially designed for persons' names (Oliveira, 2005), whereas for work and publication venue titles, we used the cosine similarity function (Salton, Wong, & Yang, 1975). In this case, we used each word in the work or publication venue title as a term and calculated the cosine between two clusters using feature vectors, where each feature corresponds to the TF-IDF (i.e., term frequency and inverse document frequency) (Baeza-Yates & Ribeiro-Neto, 1999) value of its corresponding term.

The cosine similarity function is obtained from the following formula:

$$\text{cosine}(\vec{c}_i, \vec{c}_j) = \frac{\sum_k c_{ik} * c_{jk}}{|\vec{c}_i| * |\vec{c}_j|}$$

where $\vec{c}_i$ and $\vec{c}_j$ correspond to the feature vectors of the terms in the work or publication venue titles of the clusters c_i and c_j , respectively; $|\vec{c}|$ corresponds to the norm of the vector $\vec{c}$; and c_{ik} and c_{jk} correspond to the value of the k th feature in the vectors $\vec{c}_i$ and $\vec{c}_j$, respectively.

Notice that, in our current implementation, we do not treat the case in which two similar names occur in the same citation. But, we can simply verify if there are two authorship records with the same citation identifier in two different clusters in order to not fuse them or not insert an authorship record in a cluster that has another authorship record with the same citation identifier.

Notice also that our method may form different clusters depending on the order of authorship records in the lists L and S . However, our experiments show that there is low variation on the final results. In our experiments, we randomly select the authorship records to form the lists L and S .

Comparative Evaluation

In order to evaluate our proposed method, we ran a series of experiments described below. We start by describing the collections, evaluation metrics, and baselines used in the experiments, and then present their results.

Collections

We used collections of authorship records extracted from two DLs: DBLP and BDBComp. These collections contain several ambiguous groups.

The authorship records extracted from DBLP sum up 4,287 records associated with 220 distinct authors, which means an average of approximately 20 records per author. In 2,270 of these records, the author names are in a short format corresponding to 53% of the records of the collection. The original collection was created by Han et al. (2004) and, with slight variations, it has been used in several other works (Han et al., 2004; Han, Xu, et al., 2005; Han et al., 2005; Pereira et al., 2009; Yang et al., 2008). The authors manually labeled the authorship records using external sources of information, such as the author's publication home page, curriculum vitae, affiliations, e-mail, etc. In case of doubts, they also sent e-mails to some authors to confirm the authorship of works. The records for which they had insufficient information to be judged were eliminated from the collection. Han et al. also replaced the abbreviated publication venue titles by full names obtained from DBLP. In our experiments described in the section Experiments we used the original collection with few corrections due to a small set of erroneous assignments found during the analysis of the results.

The authorship records extracted from BDBComp sum up to 363 records associated with 184 distinct authors, approximately 2 records per author. Notice that, although much smaller than the DBLP collection, this collection is very difficult to disambiguate, because it has many authors with only one record. This collection was created by us and contains the 10 largest ambiguous groups found in BDBComp in September, 2007. It is available from <http://www.lbd.dcc.ufmg.br/lbd/collections/disambiguation>.

Table 1 shows more detailed information about the collections and its ambiguous groups. Disambiguation is

particularly difficult in ambiguous groups such as the “C. Chen” group, in which the correct author must be selected from 60 possible authors, and in ambiguous groups such as the “J. Silva” group, in which the majority of authors appears in only one record. Figure 4 shows the authorship distribution within two representative groups of each collection. Notice that, few authors are very prolific and appear in several records, whereas most of the authors appear in only few records (the same trend is observed in all groups of DBLP and BDBComp). This is an intrinsic characteristic of scientific publications (Liming & Lihua, 2005).

TABLE 1. The DBLP and BDBComp collections.

DBLP		BDBComp	
Ambiguous Group	#Records/ #Authors	Ambiguous Group	#Records/ #Authors
A. Gupta	576/26	A. Oliveira	52/16
A. Kumar	243/14	A. Silva	64/32
C. Chen	798/60	F. Silva	26/20
D. Johnson	368/15	J. Oliveira	48/18
J. Martin	112/16	J. Silva	36/17
J. Robinson	171/12	J. Souza	35/11
J. Smith	921/29	L. Silva	33/18
K. Tanaka	280/10	M. Silva	21/16
M. Brown	153/13	R. Santos	20/16
M. Jones	260/13	R. Silva	28/20
M. Miller	405/12	—	—

Evaluation Metrics

To evaluate the quality of the clusters generated by each method, we used two clustering evaluation metrics: the K metric and pairwise F1. These metrics allow us to compare the methods under a number of different criteria, which is usually not done in the literature.

K metric. The K metric (Lapidot, 2002) determines the equilibrium between two specific clustering metrics: average cluster purity (ACP) and average author purity (AAP).

ACP evaluates the purity of the generated clusters with respect to reference clusters manually generated, i.e., whether the generated clusters include only records belonging to the

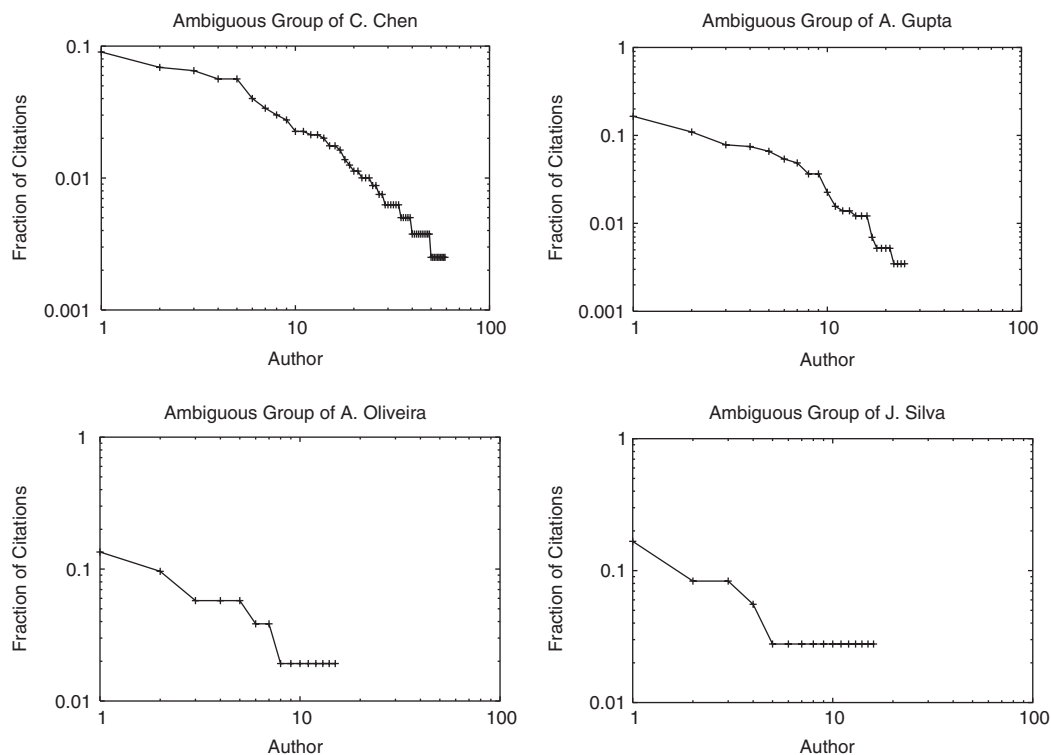


FIG. 4. Authorship distribution within four ambiguous groups. Authors (x-axis) are sorted in decreasing order of prolificity (i.e., more prolific authors appear in the first positions).

reference clusters. Thus, if the generated clusters are pure, this metric result will be 1. The formula to calculate ACP is

$$ACP = \frac{1}{N} \sum_{i=1}^q \sum_{j=1}^R \frac{n_{ij}^2}{n_i}$$

where R is the number of clusters manually generated (reference clusters); N is the total number of records in the ambiguous group; q is the number of clusters automatically generated by the method; n_{ij} is the total number of elements of cluster i automatically generated belonging to the cluster j manually generated; and n_i is the total number of items of the automatically generated cluster i .

AAP evaluates the fragmentation of automatically generated clusters with respect to the reference clusters. If there is a low proportion of fragmented clusters, the result is closer to 1. Its values vary between 0 and 1. The formula to calculate AAP is

$$AAP = \frac{1}{N} \sum_{j=1}^R \sum_{i=1}^q \frac{n_{ij}^2}{n_j}$$

where R is the number of clusters manually generated (reference clusters); N is the total number of records in the ambiguous group; q is the number of clusters automatically generated by the method; n_{ij} is the total number of elements of cluster i automatically generated belonging to the cluster j manually generated; and n_j is the total number of items of the manually generated cluster j .

The K metric consists of the geometric mean between the two metrics. It combines the evaluation of both the purity and the fragmentation of the clusters generated by each method. The formula to calculate K is:

$$K = \sqrt{ACP \times AAP}$$

Pairwise F1. Pairwise F1 (pF1) is the F1 metric (Rijsbergen, 1979) calculated using pairwise precision and pairwise recall, and measures the performance at the pair level. Pairwise precision (pP) is calculated as $pP = a/(a + c)$, where a is the number of pairwise authorship records in the same generated cluster (cluster generated by the method) that are associated with the same author and c is the number of pairwise authorship records in the same generated cluster not corresponding to the same author (Table 2). Pairwise recall (pR) is calculated as $pR = a/(a + b)$, where b is the number of pairwise authorship records associated with the same authors that are not in the same generated cluster (Table 2).

TABLE 2. Contingency table model for evaluation.

	# of pairwise records in the generated clusters	# of pairwise records not in the generated clusters
# of pairwise records of same authors	a	b
# of pairwise records of different authors	c	d

The F -metric is defined as

$$F = ((\beta^2 + 1) \times Precision \times Recall) / (\beta^2 \times Precision + Recall),$$

where β is the weight of recall relative to precision. F1 uses $\beta = 1$, which corresponds to equal weighting of the two metrics. pF1 is analogously defined and it is somehow similar to the K metric, but does not consider clusters with only one element.

Baselines

In this subsection, we give some details of the four name disambiguation methods used as baselines in our experiments: two supervised and two unsupervised methods.

The two supervised methods addressed are the ones proposed by Han et al. (2004). The first method uses the naive Bayes probability model (Mitchell, 1997) and the second uses SVMs (Cortes & Vapnik, 1995). As described in The second section, in both methods the authorship records are divided in training and testing sets. The first considered unsupervised method (Han et al., 2005) follows a clustering approach based on the K-way spectral clustering technique and the second (Han et al., 2006) uses a clustering strategy based on the DBSCAN method (Ester et al., 1996). The K-way spectral clustering-based method does not use any training data, however, it considers that the number of correct clusters is previously known. In the method by Huang et al. (2006), the DBSCAN clustering technique is used to cluster the authorship records by author after an SVM-based classifier is used to create a distance function between two records. We notice that these four methods are representative name disambiguation methods that explore several important aspects for the solution of the problem such as: (i) pairwise similarities, (ii) state-of-the-art clustering and classification algorithms, and (iii) non-linearity for cases in which ambiguous authors are not linearly separable. Moreover, these methods use the same set of citation attributes (author and coauthor names, work title and publication venue title) as evidence for the disambiguation task.

In the description below, we assume the following. Each authorship record is composed of a list of attributes that includes at least author and coauthor names, work title, and publication venue. To each attribute in a record is associated a specific instance, which may be composed of several elements. In case of the author name attribute, an element corresponds to a word of a name. Whereas in the case of the coauthor name attribute, an element corresponds to the name of a single unique person. In the case of the other mandatory attributes, elements correspond to individual keywords.

The naive Bayes method. In this method, it is assumed that each author's citation data are generated by a naive Bayes model. Thus, to estimate the model parameters some citations are used as training data.

Let X_i be an author class corresponding to a unique single person, where $i \in [1, n]$ and n is the number of authors, and

let A be an authorship record. Han et al. (2004) calculate the probability of each class X_i to generate a record A using the naive Bayes rule.

The authorship record A is attributed to a class that has the maximal posterior probability of producing it. The formula used is

$$\max_i P(X_i|A) = \max_i P(A|X_i)P(X_i)/P(A)$$

where $P(X_i)$ is the prior probability of X_i being the author of an article and $P(A)$ is the probability of A , which is omitted because it does not depend on X_i .

Thus, the formula becomes,

$$\max_i P(X_i|A) = \max_i P(A|X_i)P(X_i).$$

Assuming that attributes and distinct attribute elements are independent, the probability of the authorship record A given the author class X_i is given by

$$P(A|X_i) = \prod_j P(T_j|X_i) = \prod_j \prod_k P(T_{jk}|X_i)$$

where T_i is an attribute (T_1 the coauthor names, T_2 the work title, T_3 the publication venue title, and T_4 the author name) and each attribute has independent elements T_{jk} .

To avoid underflow, Han et al. used the log probability. Thus, the final formula becomes

$$\begin{aligned} \max_i P(X_i|A) \\ = \max_i \left[\sum_j \sum_k \log(P(T_{jk}|X_i)) + \log(P(X_i)) \right]. \end{aligned}$$

The parameters of the model for a given author X_i , which include, for example, the probability of X_i writing an article with previously seen and unseen coauthors and the probability of X_i writing a title or publishing in a venue containing a specific word, are all defined by Han et al. (2004). For probability estimation of the coauthor conditional probability $P(T_1|X_i)$, where $T_1 = (T_{11}, T_{12}, \dots, T_{1k(1)})$, Han et al. use the maximum likelihood estimation for parameters of multinomial distributions. $P(T_1|X_i)$ is decomposed into:

- $P(N|X_i)$ —the probability of X_i writing a future article alone conditioned on the event of X_i ;
- $P(Co|X_i)$ —the probability of X_i writing a future article with coauthors conditioned on the event of X_i ;
- $P(\text{Seen}|Co, X_i)$ —the probability of X_i writing a future article with previously seen coauthors conditioned on the event that X_i writes a future article with coauthors;
- $P(\text{Unseen}|Co, X_i)$ —the probability of X_i writing a future article with an unseen author conditioned on the event that X_i writes an article with coauthors;
- $P(T_{1k}|\text{Seen}, Co, X_i)$ —the probability of X_i writing a future article with a coauthor T_{1k} conditioned on event that X_i writes an article with previously seen coauthors; and
- $P(T_{1k}|\text{Unseen}, Co, X_i)$ —the probability of X_i writing an article with a coauthor T_{1k} conditioned on the event that X_i writing an article with previously unseen coauthors.

Thus, we have:

$$\begin{aligned} P(T_1|X_i) &= P(N|X_i), \text{ if there are no coauthors;} \\ P(T_1|X_i) &= P(T_{11}|X_i) \dots P(T_{1k}|X_i), \text{ if there are coauthors,} \\ &\text{where} \\ P(T_{1k}|X_i) &= P(T_{1k}, N|X_i) + P(T_{1k}, Co|X_i) \\ &= 0 + P(T_{1k}, Co|X_i) \\ &= P(T_{1k}, \text{Seen}, Co|X_i) + P(T_{1k}, \text{Unseen}, Co|X_i) \\ &= P(T_{1k}|\text{Seen}, Co, X_i) * P(\text{Seen}|Co, X_i) * P(Co|X_i) \\ &\quad + P(T_{1k}|\text{Unseen}, Co, X_i) * \\ &\quad P(\text{Unseen}|Co, X_i) * P(Co|X_i). \end{aligned}$$

The conditional probabilities $P(T_2|X_i)$, $P(T_3|X_i)$, and $P(T_4|X_i)$, which an author writes a work title, publishes in a venue and uses a name, respectively, takes each word (title or publication venue) or name independently and estimates the probability that X_i uses a particular word in a title or in a venue, or uses a name.

The SVM method. SVMs (Cortes & Vapnik, 1995) have been extensively used in classification problems. For a two-class classification problem, an SVM receives training vectors $x_i \in R^n$ (input space), $i = 1, 2, \dots, l$, and a vector $y \in \{-1, 1\}^l$, and finds an optimal linear decision hyperplane that minimizes the risk of erroneous classifications. SVMs produce a model that predicts target values of data instances in the testing set. The model is produced using a training set, which has feature vectors along with their class label that they represent. SVMs attempt to find a separated hyperplane with the maximal margin to maximally separate the classes of training sets. SVMs require both positive and negative examples to learn how to classify the data.

Han et al. (2004) associate each author name (individual person) with an author class and train the classifier for that class. Each authorship record is represented by a feature vector with the elements of their attributes (author and coauthor names, and words of work and publication venue titles) and their TFIDF as the feature weight. The authors used the “one class versus all others” approach to multi-class classification. The set of records are randomly split into training (50%) and testing (50%) sets.

The K-way spectral clustering method. In this disambiguation method, each authorship record is represented by a vertex of an undirected graph, and the weight of the edge between two vertices represents the similarity between the corresponding records. The disambiguation task consists in splitting the graph so that records that are more similar to each other belong to the same cluster.

In Han et al. (2005), each authorship record is represented as a feature vector where each feature corresponds to an element of a given instance of one of its attributes. The authors considered two options for defining the feature weights: TFIDF and Normalized Term Frequency (NTF), NTF being given by

$$\text{ntf}(i, d) = \text{freq}(i, d) / \text{maxfreq}(i, d)$$

where $\text{freq}(i, d)$ refers to the feature frequency i within the record d and $\max \text{freq}(i, d)$ refers to the maximal term frequency of feature i in the record d . In this method, we used TFIDF to be compatible with the other methods, in which we also used this similarity measure. We also noticed that in the original work the results of both options are very similar. The similarity between two authorship records is calculated using cosine similarities between the corresponding records.

The authors proposed the use of K-way spectral clustering (Zha, He, Ding, Gu, & Simon, 2001) to construct citation clusters of the same authors. To use this clustering technique, we need to inform the correct number of clusters to be generated.

The SVM-DBSCAN method. The last baseline method is the one presented in Huang et al. (2006). It attempts to solve the name ambiguity problem in two steps: first, a blocking method is applied to generate blocks of authorship records that have nonconflicting author names, reducing the number of pairwise similarity calculations; then, the DBSCAN clustering method (Ester et al., 1996), specially designed to deal with the transitivity problem, is used to create record clusters for a given author.

DBSCAN is applied to each block, using a distance function obtained from SVMs over defined vectors as a pairwise similarity vector

$$s^{(i,j)} = [\text{sim}_1(t_{u,1}^{(i)}, t_{v,1}^{(j)}), \dots, \text{sim}_m(t_{u,m}^{(i)}, t_{v,m}^{(j)})]$$

where $t_{u,k}^{(i)}$ and $t_{v,k}^{(j)}$ are pairs of attributes of authorship records of authors $a_u^{(i)}$ and $a_v^{(j)}$ in the block.

A different similarity function *can be* used for each kind of attribute. In our implementation, we used a token-based Jaccard coefficient similarity for all attributes and pairwise similarity vectors of three blocks were used for training as in Huang et al. (2006). In that work, the authors show that the number of support vectors saturates and the test error stabilizes after a certain amount of training data is used. They used an online SVM algorithm (LASVM) (Bordes, Ertekin, Weston, & Bottou, 2005) to build a supervised distance function, which yields a simpler yet faster disambiguator with active learning.

As a result of the classification, each pairwise similarity vector belongs to class 1 or class 2. These class values are used by DBSCAN as the distances between the corresponding two records, to create clusters of authorship records.

DBSCAN uses a notion of density of points, i.e., each cluster has a density of points higher than outside of the cluster. The idea is that each point of a cluster has a minimum number of neighbor points in a given radius. An interesting property of DBSCAN is that it can model clusters of any shape. To execute DBSCAN, it is necessary to provide two parameters, the minimum number of points in a cluster and the size of the epsilon neighborhood (Ester et al., 1996).

Experiments

Setup. Experiments were conducted with each ambiguous group, considering author name, list of coauthor names, work title, and publication venue title as attributes. The same HHC preprocessing was applied to all baselines, i.e., we removed punctuations and stopwords of work and publication venue titles, and applied stemming in the work titles using Porter's algorithm (Porter, 1980). Porter's algorithm removes the inflectional suffix of a word. With the use of this algorithm, we can consider two morphological variants of a word, such as "connected" and "connecting", as being represented by the same radical. Additionally, all coauthor names were standardized by considering only the initial letter of the first name along with the full last name.

We ran the experiments with each attribute (list of coauthor names, work title, and publication venue title) together with the author name and using all attributes together, as it was performed in Han et al. (2004), for the DBLP collection. This allowed us to study the contribution of each attribute in each of the methods. For the BDBComp collection, we used all the attributes. In the results shown in the following subsections, HHC represents the HHC method, NaiveBayes represents the Naive Bayes method, SVM represents the SVM method, K-way represents the K-way spectral clustering method, and SVM-DBSCAN represents the SVM-DBSCAN method.

Each method was executed 10 times. In each execution, the clustering metrics for each ambiguous group were calculated and averaged for all groups. The presented results are the mean of the averages for all runs, with the corresponding standard deviations.

For the supervised methods, each record set of an author was randomly split into training (50%) and testing (50%) sets. The other unsupervised methods were performed only with the testing sets. This split ensures a fair comparison with the unsupervised methods, including ours. All results shown below also correspond to the performance in the testing sets. The results are compared using statistical significance tests (paired t -test) with 99% confidence interval.

In order to consider the impact of the order of the authorship records on the HHC method, in each run of our experiments, the authorship records are randomly selected to form the list L with records whose author names occur in a short format and the list S with the remaining ones. This has the practical effect of altering the order of the input.

To perform the experiments with a single attribute using the HHC method we did the following. For the list of coauthor names, we executed only the first step. For work and publication venue titles, we executed the first step for creating clusters for each authorship record and then considered only one of the attributes in the second step.

In our work we used the SVM implementation provided by the LibSVM package (Chang & Lin, 2001), with radial basis function (RBF) as the kernel function, where the best γ and cost values were obtained by performing 5-fold cross-validation (Geisser, 1993) in the training data, which is a way to predict the fit of a disambiguation function to a hypothetical

TABLE 3. Results (with their respective standard deviations) using the list of coauthor names in the DBLP collection.

Method	ACP	AAP	K	pP	pR	pF1
HHC	0.98 ± 0.002	0.35 ± 0.010	0.58 ± 0.007	0.96 ± 0.007	0.22 ± 0.017	0.35 ± 0.019
SVM	0.61 ± 0.013	0.80 ± 0.006	0.70 ± 0.009	0.41 ± 0.012	0.83 ± 0.006	0.53 ± 0.011
NaiveBayes	0.68 ± 0.010	0.76 ± 0.011	0.72 ± 0.009	0.55 ± 0.016	0.77 ± 0.011	0.64 ± 0.014
K-way	0.72 ± 0.007	0.48 ± 0.010	0.59 ± 0.008	0.49 ± 0.015	0.33 ± 0.014	0.38 ± 0.014
SVM-DBSCAN	0.75 ± 0.045	0.14 ± 0.008	0.31 ± 0.004	0.11 ± 0.022	0.01 ± 0.007	0.02 ± 0.009

Note. The best results, including statistical ties, are highlighted in bold.

TABLE 4. Results (with their respective standard deviations) using the work title attribute in the DBLP collection.

Method	ACP	AAP	K	pP	pR	pF1
HHC	0.83 ± 0.007	0.33 ± 0.007	0.52 ± 0.006	0.79 ± 0.013	0.22 ± 0.008	0.34 ± 0.010
SVM	0.65 ± 0.010	0.79 ± 0.009	0.72 ± 0.009	0.50 ± 0.013	0.85 ± 0.005	0.61 ± 0.011
NaiveBayes	0.43 ± 0.011	0.78 ± 0.009	0.57 ± 0.009	0.30 ± 0.007	0.80 ± 0.007	0.43 ± 0.008
K-way	0.67 ± 0.012	0.42 ± 0.008	0.53 ± 0.009	0.54 ± 0.028	0.26 ± 0.008	0.34 ± 0.012
SVM-DBSCAN	0.23 ± 0.036	0.85 ± 0.085	0.43 ± 0.011	0.17 ± 0.004	0.80 ± 0.100	0.27 ± 0.010

Note. The best results, including statistical ties, are highlighted in bold.

TABLE 5. Results (with their respective standard deviations) using the publication venue title attribute in the DBLP collection.

Method	ACP	AAP	K	pP	pR	pF1
HHC	0.82 ± 0.012	0.26 ± 0.005	0.46 ± 0.007	0.64 ± 0.023	0.15 ± 0.006	0.23 ± 0.007
SVM	0.60 ± 0.010	0.76 ± 0.007	0.67 ± 0.008	0.43 ± 0.014	0.81 ± 0.009	0.55 ± 0.012
NaiveBayes	0.36 ± 0.007	0.77 ± 0.005	0.52 ± 0.006	0.26 ± 0.005	0.77 ± 0.006	0.38 ± 0.007
K-way	0.62 ± 0.009	0.39 ± 0.010	0.49 ± 0.010	0.49 ± 0.015	0.25 ± 0.011	0.32 ± 0.011
SVM-DBSCAN	0.83 ± 0.031	0.13 ± 0.003	0.31 ± 0.003	0.05 ± 0.044	0.00 ± 0.002	0.01 ± 0.003

Note. The best results, including statistical ties, are highlighted in bold.

validation set. In our experiments, we used TFIDF as the feature weight scheme. For the K-way method, we used the implementation of the K-way spectral clustering provided by the University of Washington spectral clustering working group.⁷

For the SVM-DBSCAN method, we used the LibSVM package with the values 2,048 and 0.00048828125 for cost and γ , respectively, also obtained with cross-validation in the training data, and the DBSCAN version available from Weka⁸ with the values of 1.2 and 8 for ϵ and the minimum number of points, respectively. We also used the Fragment Comparison algorithm (Oliveira, 2005) for determining the similarity between author and coauthor names, and the token-based Jaccard⁹ coefficient as the similarity measure for work and publication venuetitles when constructing the similarity vectors. In addition, we used the similarity vectors of “A Kumar,” “D Johnson,” and “M Jones,” ambiguous groups as training sets for the SVM classifier, as done in Huang et al. (2006). For the HHC method, we used the values of 0.2 and 0.4 for similarity thresholds of venues and titles, respectively,

on the DBLP collection, and the value of 0.3 for both similarity thresholds on the BDBComp collection. These values were the ones that produced the best empirical results.

Results. Table 3 shows the averaged results for the experiments using only the list of coauthor names in the DBLP collection.

As explained before, this experiment corresponds to the first step of the HHC method. We notice that HHC produces very fragmented clusters (AAP = 0.35 ± 0.010 and pR = 0.22 ± 0.017) using only the list of coauthor names. However, the produced clusters are very pure (ACP = 0.98 ± 0.002 and pP = 0.96 ± 0.007). We also notice that the supervised methods produced the best results using only the list of coauthor names under all metrics.

Table 4 shows the averaged results for the experiments using only the work title attribute. As it happened with the list of coauthor names, we notice that HHC produces very fragmented clusters (AAP = 0.33 ± 0.007 and pR = 0.22 ± 0.008). However, the produced clusters are the purest (ACP = 0.83 ± 0.007 and pP = 0.79 ± 0.014).

Table 5 shows the averaged results for the experiments using only the publication venue title attribute. HHC produces again pure (ACP = 0.82 ± 0.012 and pP = 0.64 ± 0.023), but very fragmented clusters (AAP = 0.26 ± 0.005

⁷<http://www.stat.washington.edu/spectral>

⁸<http://www.cs.waikato.ac.nz/ml/weka/>

⁹The implementation of the Jaccard coefficient in our work is the one available at <http://www.dcs.shef.ac.uk/~sam/simmetrics.html>

TABLE 6. Results (with their respective standard deviations) using all attributes together in the DBLP collection.

Method	ACP	AAP	K	pP	pR	pF1
HHC	0.86 ± 0.010	0.68 ± 0.011	0.77 ± 0.008	0.84 ± 0.014	0.65 ± 0.017	0.73 ± 0.013
SVM	0.75 ± 0.010	0.85 ± 0.006	0.80 ± 0.008	0.61 ± 0.012	0.91 ± 0.007	0.72 ± 0.010
NaiveBayes	0.67 ± 0.011	0.80 ± 0.009	0.73 ± 0.009	0.53 ± 0.011	0.85 ± 0.009	0.64 ± 0.010
K-way	0.75 ± 0.011	0.47 ± 0.009	0.59 ± 0.009	0.66 ± 0.017	0.30 ± 0.008	0.40 ± 0.010
SVM-DBSCAN	0.24 ± 0.039	0.83 ± 0.082	0.43 ± 0.013	0.17 ± 0.007	0.78 ± 0.092	0.27 ± 0.010

Note. The best results, including statistical ties, are highlighted in bold.

TABLE 7. Results (with their respective standard deviations) for the HHC and SVM-based methods for each ambiguous group of the DBLP collection.

Ambiguous Group	HHC			SVM		
	ACP	AAP	K	ACP	AAP	K
A. Gupta	0.87 ± 0.026	0.72 ± 0.043	0.79 ± 0.031	0.79 ± 0.019	0.87 ± 0.014	0.83 ± 0.016
A. Kumar	0.87 ± 0.041	0.65 ± 0.052	0.75 ± 0.042	0.72 ± 0.039	0.83 ± 0.032	0.77 ± 0.034
C. Chen	0.55 ± 0.017	0.53 ± 0.017	0.54 ± 0.007	0.66 ± 0.027	0.76 ± 0.021	0.71 ± 0.023
D. Johnson	0.86 ± 0.027	0.60 ± 0.036	0.72 ± 0.023	0.69 ± 0.047	0.81 ± 0.023	0.75 ± 0.034
J. Martin	0.94 ± 0.022	0.84 ± 0.051	0.89 ± 0.035	0.63 ± 0.050	0.85 ± 0.025	0.73 ± 0.032
J. Robinson	0.91 ± 0.033	0.65 ± 0.041	0.76 ± 0.026	0.74 ± 0.048	0.84 ± 0.035	0.79 ± 0.034
J. Smith	0.82 ± 0.025	0.61 ± 0.034	0.71 ± 0.028	0.82 ± 0.018	0.88 ± 0.010	0.85 ± 0.013
K. Tanaka	0.90 ± 0.023	0.73 ± 0.037	0.81 ± 0.021	0.85 ± 0.028	0.91 ± 0.022	0.88 ± 0.023
M. Brown	0.91 ± 0.031	0.76 ± 0.063	0.83 ± 0.040	0.73 ± 0.059	0.84 ± 0.030	0.79 ± 0.044
M. Jones	0.91 ± 0.021	0.64 ± 0.024	0.76 ± 0.018	0.74 ± 0.046	0.85 ± 0.031	0.79 ± 0.036
M. Miller	0.95 ± 0.013	0.81 ± 0.033	0.87 ± 0.022	0.85 ± 0.019	0.93 ± 0.013	0.89 ± 0.015

Note. The best results, including statistical ties, are highlighted in bold.

and $pR = 0.15 \pm 0.006$) as it happened with the list of coauthor names and work title attributes.

Table 6 shows the averaged results for the experiments using all attributes together. In this case, HHC obtains high purity and low fragmentation. As we can see, HHC outperforms all the four methods under the two considered metrics (K , $pF1$), except the SVM method for which there is a statistical tie under the $pF1$ metric and a very, close result under K metric. However, we should recall that HHC uses no training data, whereas the SVM method does, which is a major advantage for HHC from a practical viewpoint. The good performance of the HHC method can be attributed to its specialized heuristics which consider that very rarely two authors with similar names and sharing a (similar) coauthor in common would be two different persons in the real world and that authors tend to publish in the same subjects and publication venues for some portion of their careers.

On the other hand, the poor performance of the SVM-DBSCAN may be attributed to the few attributes that we normally have in the particular scenario we are focusing on, i.e., author name disambiguation in bibliographic citations. In the original scenario in which this method was proposed (Huang et al., 2006), the authors used many more attributes and features, as it is also done in other works (Torvik & Smalheiser, 2009; Torvik et al., 2005; Treeratpituk & Giles, 2009).

To better assess the relative difficulty of the disambiguation task in the discussed experiments, we considered one

run of the three best methods (SVM, Naive Bayes, and HHC) applied to the same test set and verified the amount of errors in common among them in the DBLP collection. In the case of HHC, since it is a clustering method, we sorted the output clusters by the amount of records and we defined the class (author) of the cluster as the most frequent class in that cluster. If the most frequent class had already been assigned to another cluster, we used the second most frequent one and so on. And if all the classes present in the cluster had already been assigned to some other cluster, a new class was generated, i.e., an author not present in the training set was assigned to that cluster, which means, in practical terms for evaluation purposes, that all records in that cluster were wrongly assigned. As a result, in that run, 118 out of 2,081 records were erroneously classified by all three methods (approximately 5.6% of the test set), whereas 217 records (approximately 10.4% of the test set) were erroneously classified by the SVM and NaiveBayes. This shows that there are really difficult cases in the collections that no method can correctly disambiguate.

Table 7 shows the results for each ambiguous group in the DBLP collection for the HHC and SVM methods, the overall best two methods, under the ACP, AAP, and K metrics. We can notice that there were statistical ties in 6 out of 11 groups, with one win for HHC, and that when the HHC had worse performance, the results were close to those of the SVM, except for the cases of the groups JSmith and CChen. We also notice that the purity of the clusters produced by HHC

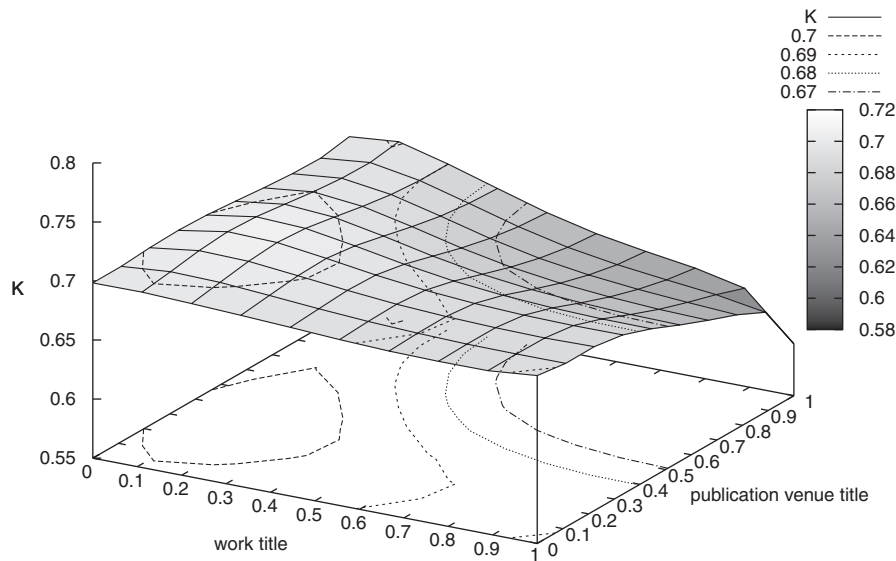


FIG. 5. Sensitivity to thresholds of the work and publication venue titles.

TABLE 8. Results (with their respective standard deviations) using all attributes together on BDBComp collection.

Method	ACP	AAP	K	pP	pR	$pF1$
HHC	0.88 ± 0.021	0.99 ± 0.010	0.93 ± 0.015	0.58 ± 0.085	0.83 ± 0.119	0.65 ± 0.089
SVM	0.26 ± 0.028	0.95 ± 0.018	0.48 ± 0.024	0.10 ± 0.025	0.70 ± 0.136	0.16 ± 0.032
Naive Bayes	0.20 ± 0.008	0.97 ± 0.020	0.42 ± 0.009	0.10 ± 0.016	0.80 ± 0.131	0.16 ± 0.019
K-way	0.89 ± 0.017	0.97 ± 0.016	0.93 ± 0.015	0.67 ± 0.122	0.79 ± 0.140	0.71 ± 0.129
SVM-DBSCAN	0.36 ± 0.117	0.80 ± 0.066	0.48 ± 0.069	0.04 ± 0.023	0.31 ± 0.215	0.05 ± 0.028

Note. The best results, including statistical ties, are highlighted in bold.

is higher, except, again, for the CChen group. This suggests that we should have a specific strategy for Chinese and Asian names in general.

We next analyze the sensitivity of HHC to the similarity thresholds of the work and publication venue titles using the DBLP collection. Figure 5 shows the results of this analysis in which we varied the threshold values of the work and publication venue titles from 0.0 to 1.0 and for each pair of thresholds (work and publication venue titles). We performed 10 executions of the methods and the results reported correspond to the average of all executions. We can notice that, when the threshold value is 0.0 for both attributes, the K value is close to 0.70. When the similarity threshold of the publication venue titles increases up to 0.4, the K values also increase. After this, the K values begin to decrease. When the similarity threshold of the work titles increases up to 0.2, the K values also increase. After this, the K values start to decrease. The highest K value is obtained when the thresholds of the work and publication venue titles are 0.2 and 0.4, respectively.

Table 8 shows the averaged results for the methods applied to the BDBComp collection with all attributes. The main characteristic of this collection is the large number of authors (approximately 73%) with only one authorship record. We can notice that, under K and $pF1$ metrics, HHC and K-way produced the best results (both with a statistical tie). However,

it is important to notice that the K-way method exploits the exact number of groups to be generated in its execution while HHC does not. In practice, it is not realistic to assume that we may know how many groups should be generated a priori. The poor performance of the supervised methods is probably due to their inability to generalize in the presence of small amounts of training data, which are, in many practical cases, very difficult to obtain.

Analysis of Cases of Failure

In order to explain some results and to further illustrate the difficulty of the problem, in Table 8 we show four examples in which HHC has failed to disambiguate specific authorship records. The first example (Failure 1) is a case of failure in the method's first step. Two authors of the same ambiguous group ("C. Chen") were considered the same author. The c_1 and c_2 records include similar author names ("Chuen Liang Chen" and "C. Chen") and similar coauthor names ("Biing Feng Wang" and "B. Wang"). In this case, the records were assigned to the same author, when in fact they belonged to different authors.

In the second case of failure (Failure 2), the citation records were erroneously grouped in the ambiguous group "A. Gupta." Both records were grouped together because they refer to authors with similar names and that have published in

TABLE 9. Illustrative examples in which HHC failed to disambiguate.

Failure	Label	Authors	Work Title	Publication Venue Title
1	c ₁	Chuen-Liang Chen, Biing-Feng Wang, Gen-Huey Chen	A Simple Approach to Implementing Multiplication with Small Tables	Inf. Process. Lett.
	c ₂	C. Chen, Yang Xiao, B. Wang	Bandwidth Degradation QoS for Adaptive Multimedia in Wireless/Mobile Networks	Computer Communications Elsevier Science
2	c ₃	Amarnath Gupta, Yang Yang, Aditya Bagchi, Animesh Ray	Declarative Specification of Gene Regulatory Networks and Query Evaluation in PathSys	International Conference on Data Engineering
	c ₄	A. Gupta, Manish Bhide, Mukesh Mohania	Towards Bringing Database Management Task Realm It Non Experts	International Conference on Data Engineering
3	c ₅	A. Gupta, Chris Nolt, Jaswinder Pal Singh, John L Hennessy, Takashi Totsuka	Load Balancing Locality Adaptive Hierarchical N Body Methods Barnes Hut Fast Multipole Rasiosity	J Parallel Distrib Comput
	c ₆	Ajay Gupta, Elise De Doncker, Patricia Ealy	Two Methods Load Balanced Distributed Adaptive Integration	European International Conference High Performance Computing And Networking Europe
4	c ₇	Amar Gupta, B. E. Prasad, M. P. Reddy, P. G. Reddy	Methodology Integration Heterogeneous Databases	IEEE Trans. Knowl. Data Eng.
	c ₈	A. Gupta, A. Liu, M. V. Nagendraprasad, Patrick Shen Pei Wang, S. Ayyadurai	Integrated Architecture Recognition Totally Unconstrained Handwritten Numerals	International Journal of Pattern Recognition and Artificial Intelligence

the same venue, the International Conference on Data Engineering (ICDE). However, the first record refers to “Amarnath Gupta,” whereas the second refers to “Ajay Gupta.”

The third case illustrates a failure that occurs when the authorship records include similar author names and there are several (common) words in the corresponding work titles. In this case, both records include similar author names (“A. Gupta” and “Ajay Gupta”) and four title words in common (“load,” “balanc,” “adaptiv,” and “method”).

The last example of failure (Failure 4) shows two authorship records of the same author (“Amar Gupta”, professor of the Eller College of Management, University of Arizona). Those authorship records were assigned to distinct groups because their list of coauthor names do not include any similar name, their work titles have only one term in common, and their publication venue titles have no terms in common.

Conclusion

In this article, we presented a HHC method that uses specific heuristics to solve the author name ambiguity problem in bibliographic citations. This method also deals with both the SC and MC problems, which is a major advantage when compared with other existing methods in the literature such as Lee et al. (2005) and On et al. (2005).

Experiments were performed with four representative baseline methods using the same common collections, with the same set of bibliographic attributes (author and coauthor names, work title, and publication venue title), and under two evaluation metrics. The results were presented for each attribute in isolation and for the combination of them, and were statistically analyzed.

Using the DBLP collection, considering all attributes together, the best results under both metrics were produced by the HHC method except when we compared it with the SVM method under the K measure in which the HHC results

are very close to the SVM results, without the burden of any training. The good performance of the HHC method can be attributed to its heuristics, which consider that, very rarely, two authors with similar names and sharing a coauthor in common would be two different persons in the real world and that authors tend to publish in the same subjects and publication venues for some portion of their careers.

Using the BDBComp collection, the HHC and K-way methods produced the best results under both metrics. In addition, HHC does not make use of any privileged information such as the number of correct groups to be generated, as the K-way method does.

We should stress that our method is specially suitable for disambiguating author names in the context of bibliographic citations with only few attributes (in our case, the three most common attributes) or when there is not enough data to train a supervised disambiguator. Bibliographic citations, in the sense that we used in our work, play an important role in many DLs besides DBLP and BDBComp, such as CiteSeer, arXiv, e-Prints, Google Scholar, Microsoft Libra, the Collection of Computer Science Bibliographies, etc. Citations are also one of the most common types of information found in articles, reports, curricula vitae, etc., and are the basis for several interoperability solutions such as those from the Open Archives Initiative (Lagoze & de Sompel, 2001). This shows the potential of the practical application of our method. In situations in which there are more attributes or more generic feature sets and abundant training data available, supervised methods such as SVMs have the potential to work better, but ours can also be extended to deal with more attributes by properly including new similarity measures for the new features. We leave this for future work.

Another great benefit of our method is that it can, almost without modification, perform online clustering of updates as it merges pairs of clusters that are similar (i.e., have a similarity above a fixed threshold) as they are encountered.

In this scenario, the method could also be extended to find the “best merging” when there are multiple similar clusters in order to improve the quality of the clusters at the cost of some speed. This is also for future work.

Extensions of our work can also benefit and complement the current efforts to create author IDs by disambiguating legacy collections of bibliographic citations, associating existing authors in the collections with some canonical names and, in the future, with their author IDs.

Also as future work, we intend to explore classification strategies (e.g., to classify records under Computer Science sub-areas) in order to generate new pieces of evidence to improve our method. We also want to exploit other alternatives for evidence combination (e.g., genetic programming, association rules) to be used in variations of our method. Finally, we want to investigate the use of our method in a real-world scenario of constant updates of the bibliographic repositories, since this scenario imposes a major challenge to supervised methods which may need new training data for updating the learned models to deal with new incoming authors.

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